

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

Software Engineer — HCL Technologies, Chennai, India

Sep 2021 – Jan 2025

- Developed and executed process validation protocols (IQ/OQ/PQ) for precision braiding machinery, ensuring adherence to **ISO 13485** and **FDA QMSR** standards.
- Operated and maintained advanced wire braiding machines, including **Steeger** and **Herzog** models, to produce implantable components with tight engineering tolerances.
- Designed and implemented test method validation procedures for metrology tools, such as **micrometers** and **vision systems**, to verify product specifications.
- Collaborated with cross-functional teams of scientists and engineers to align project goals with customer regulatory and quality standards, contributing to process improvements.
- Generated comprehensive technical documentation and test reports detailing experimental results and process evaluations, ensuring regulatory compliance and IP protection.
- Managed assigned projects, including timelines and resource allocation, to ensure successful delivery of high-precision braiding operations using **Nitinol wire** and polymer filaments.
- Contributed to risk management activities, including the creation and execution of **pFMEA** analyses, to identify and mitigate potential process failures.

SKILLS

Manufacturing – Wire Braiding Machines, Steeger, Herzog, Nitinol Processing, Implantable Components

Quality & Regulatory – ISO 13485, FDA QMSR, Process Validation, IQ/OQ/PQ, Risk Management, pFMEA

Metrology & Design – Precision Metrology Tools, Micrometers, Vision Systems, Tensile Testers, 3D CAD Modelling, Geometric Dimensioning and Tolerancing (GD&T;)

Engineering & Analysis – Process Design, Data Analysis, Process Optimization, Prototyping, Test Method Design, Technical Documentation, Report Writing

Software & Tools – Jira, Confluence, Python, SQL, Java, Spring Boot, AWS, Docker, Linux, Git, Jenkins, Splunk, Dynatrace, Power BI, Tableau, R, pandas, NumPy, scikit-learn

PROJECTS

Nitinol Braiding Process Optimization (HCL Technologies)

- Engineered and optimized a Python-based data analysis pipeline to process sensor data from braiding machinery, identifying key parameters affecting wire tension and uniformity.
- Developed and validated new process parameters for braiding Nitinol wire using advanced machinery, resulting in a **15% reduction** in material defects observed during quality checks.
- Documented the refined braiding process, including operating procedures and quality control checkpoints, for integration into the broader manufacturing workflow.

Medical Device Component Prototyping (MSc Data Analytics Project)

- Designed and prototyped novel implantable component geometries using 3D CAD software, translating clinical requirements into manufacturable designs.
- Conducted material property analysis on various polymer filaments and **Nitinol** alloys, performing statistical analysis in R to select optimal materials for specific device applications.
- Developed a Power BI dashboard to visualize prototype performance metrics and material characteristics, facilitating design review discussions with project stakeholders.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin

Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India

2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)